

Multimedia Databases Exercises

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Exercise 9

Subject: Indexing
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Task 1: Index structures in general

1. What are the general functions of index structures in databases?
2. Which other requirements does CBIR have on index structures?
3. Among the frequently-used index structures, we list the **B-tree**, **Hash**, **K-d tree** and **Point Quadtree**. For each one, explain its structure and its functioning.

Task 2: Index structures for n-dimensional data: R-tree

1. Describe the most important characteristics and the structure of an R-tree.
2. Explain how the functions of inset and delete work in an R-tree-index.

Task 3: Search algorithms for n-dimensional objects in databases

Describe the following algorithms for similarity search with and without index:

1. Exact queries
2. Range queries
3. Nearest neighbour queries